12TH CHEMISTRY · CHAPTER 27 · FULL CHAPTER MCQS

Biochemistry

151 topic-wise MCQs from the complete chapter with answers and reasons

27.1 CARBOHYDRATES

1. The study of chemical substances found in living organisms and their reactions is:

(a) electrochemistry (b) geochemistry
(c) biochemistry (d) photochemistry

Answer: (c) biochemistry — Definition.

2. Chemically, carbohydrates are:

(a) polymers of nucleotides (b) esters of fatty acids
(c) polymers of amino acids (d) polyhydroxy aldehydes or ketones

Answer: (d) polyhydroxy aldehydes or ketones — Or their derived compounds.

3. The simplest carbohydrates are:

(a) monosaccharides (b) disaccharides
(c) polysaccharides (d) lipids

Answer: (a) monosaccharides — They cannot be hydrolysed further.

4. In the general formula of monosaccharides, n ranges from:

(a) one to three (b) three to seven
(c) ten to twenty (d) six to twelve

Answer: (b) three to seven — Triose to heptose.

5. A monosaccharide with five carbon atoms is a:

(a) hexose (b) triose
(c) tetrose (d) pentose

Answer: (d) pentose — Named by the number of carbons.

6. A monosaccharide with six carbon atoms is a:

(a) hexose (b) pentose
(c) triose (d) tetrose

Answer: (a) hexose — E.g. glucose.

7. Glucose is an:

(a) amino acid (b) ketose
(c) aldose (d) ester

Answer: (c) aldose — It has an aldehydic group.

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8. Fructose is a:

(a) aldose (b) ketose
(c) disaccharide (d) polysaccharide

Answer: (b) ketose — It has a ketonic group.

9. The six-membered ring of glucose is called:

(a) pyranose (b) furanose
(c) purine (d) pyrimidine

Answer: (a) pyranose — Fructose forms a furanose ring.

10. The five-membered ring of fructose is called:

(a) pyranose (b) furanose
(c) steroid (d) helix

Answer: (b) furanose — Glucose forms a pyranose ring.

11. Monosaccharides are solids in pure form because of:

(a) metallic bonding (b) ionic bonding
(c) extensive hydrogen bonding (d) disulphide bridges

Answer: (c) extensive hydrogen bonding — Due to their many –OH groups.

12. Monosaccharides are highly soluble in water because they:

(a) contain long hydrocarbon chains (b) are non-polar
(c) are ionic salts (d) form hydrogen bonds with water

Answer: (d) form hydrogen bonds with water — Through their –OH groups.

13. The sugar called blood sugar is:

(a) glucose (b) fructose
(c) sucrose (d) lactose

Answer: (a) glucose — It is present in human blood.

14. The sugar abundant in fruits and honey is:

(a) maltose (b) lactose
(c) fructose (d) cellulose

Answer: (c) fructose — Stated in the book.

15. The viscosity of honey is due to:

(a) ionic bonds (b) peptide bonds
(c) phosphodiester bonds (d) strong hydrogen bonding

Answer: (d) strong hydrogen bonding — Between fructose and glucose molecules.

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16. The general formula of common disaccharides is:

(a) $C_6H_{12}O_6$
(b) $C_{12}H_{22}O_{11}$
(c) $C_6H_{10}O_5$
(d) $C_{12}H_{24}O_{12}$

Answer: (b) $C_{12}H_{22}O_{11}$ — Two hexoses minus one water.

17. The C–O–C bond joining two monosaccharides is the:

(a) ester bond
(b) peptide bond
(c) glycosidic bond
(d) phosphodiester bond

Answer: (c) glycosidic bond — Formed by condensation.

18. Sucrose is made of:

(a) glucose and fructose
(b) glucose and glucose
(c) glucose and galactose
(d) fructose and fructose

Answer: (a) glucose and fructose — Common table sugar.

19. When a glycosidic bond forms, the molecule released is:

(a) carbon dioxide
(b) water
(c) ammonia
(d) hydrogen

Answer: (b) water — It is a condensation reaction.

20. Which one is a disaccharide?

(a) glycogen
(b) glucose
(c) starch
(d) maltose

Answer: (d) maltose — Sucrose, lactose and maltose are disaccharides.

21. The general formula of polysaccharides is:

(a) $C_6H_{12}O_6$
(b) $(C_6H_{10}O_5)_n$
(c) $C_{12}H_{22}O_{11}$
(d) $(CH_2)_n$

Answer: (b) $(C_6H_{10}O_5)_n$ — Given in the book.

22. The main energy store of plants is:

(a) starch
(b) cellulose
(c) glycogen
(d) pectin

Answer: (a) starch — It consists of glucose units.

23. The two components of starch are:

(a) glycogen and amylose
(b) glucose and fructose
(c) cellulose and pectin
(d) amylose and amylopectin

Answer: (d) amylose and amylopectin — Unbranched and branched.

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24. Amylose has:

(a) five-membered rings only (b) highly branched chains
(c) long unbranched chains (d) peptide links

Answer: (c) long unbranched chains — With many glycosidic linkages.

25. Amylopectin is:

(a) unbranched (b) compact and highly branched
(c) a disaccharide (d) a protein

Answer: (b) compact and highly branched — Many glucose units.

26. The major structural material of plant cell walls is:

(a) cellulose (b) starch
(c) glycogen (d) sucrose

Answer: (a) cellulose — Long unbranched chains of glucose.

27. Cellulose chains contain up to about:

(a) 250 000 fructose units (b) 25 glucose units
(c) 2 glucose units (d) 2500 glucose units

Answer: (d) 2500 glucose units — Stated in the book.

28. Cellulose in human food acts as:

(a) a hormone (b) an enzyme
(c) fibre (d) a vitamin

Answer: (c) fibre — It improves gut health.

29. Glycogen is stored in the:

(a) blood plasma only (b) bones and teeth
(c) skin and hair (d) liver and muscles

Answer: (d) liver and muscles — It is ‘animal starch’.

30. Glycogen resembles amylopectin but is:

(a) made of fructose (b) unbranched
(c) more highly branched (d) a disaccharide

Answer: (c) more highly branched — This makes it different from starch.

31. One gram of glucose provides about:

(a) 15.6 kJ (b) 1.56 kJ
(c) 156 kJ (d) 37 kJ

Answer: (a) 15.6 kJ — Stated in the book.

32. The normal blood glucose level is:

(a) 7–11 mg/dL (b) 70–110 mg/dL
(c) 170–210 mg/dL (d) 700–1100 mg/dL

Answer: (b) 70–110 mg/dL — Controlled by insulin and glucagon.

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33. Blood glucose is controlled by insulin and:

(a) glucagon (b) keratin
(c) ferritin (d) collagen

Answer: (a) glucagon — Both are hormones.

34. High blood sugar is called:

(a) hydrolysis (b) hemolysis
(c) hibernation (d) hyperglycemia

Answer: (d) hyperglycemia — Seen in uncontrolled diabetes.

35. A high intake of fructose causes:

(a) tooth decay only (b) anaemia
(c) obesity (d) night blindness

Answer: (c) obesity — Excess fat accumulates in the body.

27.2 PROTEINS

36. Complete hydrolysis of proteins gives:

(a) nucleotides (b) fatty acids
(c) monosaccharides (d) amino acids

Answer: (d) amino acids — Proteins are chains of amino acids.

37. The number of different amino acids that make proteins is about:

(a) 4 (b) 20
(c) 64 (d) 100

Answer: (b) 20 — Stated in the book.

38. A peptide bond is:

(a) an amide bond (b) an ester bond
(c) an ether bond (d) a glycosidic bond

Answer: (a) an amide bond — –CO–NH–.

39. Long chains of amino acids are called:

(a) polynucleotides (b) polysaccharides
(c) polypeptide chains (d) triglycerides

Answer: (c) polypeptide chains — Formed by many peptide linkages.

40. The specific linear sequence of amino acids is the:

(a) primary structure (b) secondary structure
(c) tertiary structure (d) quaternary structure

Answer: (a) primary structure — Held by peptide bonds.

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41. Sickle cell anaemia is caused by abnormal:

(a) collagen (b) insulin
(c) keratin (d) haemoglobin

Answer: (d) haemoglobin — Red cells become sickle shaped.

42. Albumin contains about:

(a) 40 amino acids (b) 4 amino acids
(c) 400 amino acids (d) 4000 amino acids

Answer: (c) 400 amino acids — Stated in the book.

43. α-Helices and β-pleated sheets belong to the:

(a) primary structure (b) secondary structure
(c) tertiary structure (d) quaternary structure

Answer: (b) secondary structure — Formed by hydrogen bonds of the backbone.

44. The secondary structure is held by:

(a) phosphodiester bonds (b) peptide bonds only
(c) glycosidic bonds (d) hydrogen bonds

Answer: (d) hydrogen bonds — Within or between chains.

45. An example of a protein with α-helix structure is:

(a) starch (b) silk fibroin
(c) keratin (d) cholesterol

Answer: (c) keratin — A coiled, spiral structure.

46. An example of a protein with β-pleated sheets is:

(a) silk fibroin (b) keratin
(c) insulin (d) glycogen

Answer: (a) silk fibroin — A folded, sheet-like structure.

47. The overall three-dimensional folding of one polypeptide chain is the:

(a) primary structure (b) tertiary structure
(c) secondary structure (d) quaternary structure

Answer: (b) tertiary structure — Due to interactions between R-groups.

48. Which is NOT an interaction of the tertiary structure?

(a) glycosidic bonds (b) hydrogen bonds
(c) disulphide bridges (d) hydrophobic interactions

Answer: (a) glycosidic bonds — Glycosidic bonds occur in carbohydrates.

49. The association of two or more folded polypeptide chains gives the:

(a) secondary structure (b) primary structure
(c) quaternary structure (d) tertiary structure

Answer: (c) quaternary structure — The chains are called subunits.

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50. Fibrous proteins are generally:

(a) enzymes (b) soluble in water
(c) spherical (d) insoluble in water

Answer: (d) insoluble in water — They are strong and stable.

51. Which one is a fibrous protein?

(a) insulin (b) collagen
(c) egg albumin (d) an enzyme

Answer: (b) collagen — Found in connective tissue.

52. Most enzymes and hormones are:

(a) polysaccharides (b) fibrous proteins
(c) lipids (d) globular proteins

Answer: (d) globular proteins — They are the worker proteins.

53. Globular proteins are more sensitive to:

(a) heat and pH (b) light only
(c) sound (d) pressure only

Answer: (a) heat and pH — They are easily denatured.

54. Egg white hardens on heating because:

(a) water is absorbed
(b) peptide bonds are formed
(c) hydrogen bonds between chains are disrupted
(d) it is oxidised to fat

Answer: (c) hydrogen bonds between chains are disrupted — The chains open up (denaturing).

55. Milk protein precipitated by lemon juice is:

(a) keratin (b) casein
(c) collagen (d) ferritin

Answer: (b) casein — The pH is lowered.

56. Insulin is produced by the:

(a) kidney (b) liver
(c) pancreas (d) stomach

Answer: (c) pancreas — It is a hormonal protein.

57. The protein which stores iron is:

(a) ferritin (b) fibrin
(c) actin (d) keratin

Answer: (a) ferritin — A storage protein.

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58. Fibrin is found in:

(a) egg white (b) hair
(c) muscle filaments (d) blood clots

Answer: (d) blood clots — A structural protein.

59. Actin and myosin are:

(a) storage proteins (b) contractile proteins
(c) defensive proteins (d) hormonal proteins

Answer: (b) contractile proteins — They form the filaments of muscle.

60. Antibodies are:

(a) storage proteins (b) structural proteins
(c) defensive proteins (d) contractile proteins

Answer: (c) defensive proteins — They bind to specific antigens.

61. Haemoglobin is a:

(a) contractile protein (b) structural protein
(c) receptor protein (d) transport protein

Answer: (d) transport protein — Its iron atom carries oxygen.

62. Receptor proteins are located in the:

(a) nucleus only (b) cell membrane
(c) bone matrix (d) blood plasma

Answer: (b) cell membrane — They receive chemical signals.

63. Keratin is found in:

(a) hair and nails (b) blood plasma
(c) egg white (d) milk

Answer: (a) hair and nails — A structural, fibrous protein.

27.3 ENZYMES AS BIOCATALYSTS

64. Enzymes are:

(a) lipids that store energy
(b) proteins that act as biological catalysts
(c) carbohydrates
(d) minerals

Answer: (b) proteins that act as biological catalysts — Definition.

65. One molecule of carbonic anhydrase converts in one minute about:

(a) one molecule (b) one hundred molecules
(c) ten molecules (d) one million molecules

Answer: (d) one million molecules — CO_2 and H_2O into carbonic acid.

66. The site of an enzyme which binds the substrate is the:

(a) backbone (b) nucleus
(c) active site (d) peptide bond

Answer: (c) active site — It converts the substrate into products.

67. The temporary combination of enzyme and substrate is the:

(a) enzyme–substrate complex (b) inhibitor complex
(c) co-enzyme (d) polypeptide

Answer: (a) enzyme–substrate complex — The products are then released.

68. After the reaction, the enzyme is:

(a) denatured (b) used up
(c) unchanged (d) converted into product

Answer: (c) unchanged — It is free to bind another substrate.

69. The lock and key model was proposed by:

(a) Woodward (b) Watson and Crick
(c) Karl Ziegler (d) Emil Fischer

Answer: (d) Emil Fischer — In 1894.

70. In the lock and key model, the key is the:

(a) enzyme (b) substrate
(c) product (d) inhibitor

Answer: (b) substrate — The lock is the enzyme.

71. In the lock and key model, the keyhole is the:

(a) active site (b) substrate
(c) product (d) co-enzyme

Answer: (a) active site — Only the correct key fits.

72. The lock and key model was proposed in:

(a) 1972 (b) 1953
(c) 1894 (d) 1828

Answer: (c) 1894 — By Emil Fischer.

73. The temperature giving maximum enzyme activity is the:

(a) boiling point (b) optimum temperature
(c) melting point (d) critical temperature

Answer: (b) optimum temperature — Similarly for pH.

74. Human enzymes work best at about:

(a) 73 °C (b) 0 °C
(c) 100 °C (d) 37 °C

Answer: (d) 37 °C — Body temperature.

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75. The pH range in which most human enzymatic reactions occur best is:

(a) 6.4–6.9 (b) 1.0–2.0
(c) 9.0–10.0 (d) 12–14

Answer: (a) 6.4–6.9 — Stated in the book.

76. Increasing the enzyme or substrate concentration:

(a) stops the reaction (b) increases the rate
(c) denatures the enzyme (d) has no effect at all

Answer: (b) increases the rate — More enzyme–substrate complex forms.

77. Substances which make enzymes lose their catalytic activity are:

(a) substrates (b) activators
(c) co-enzymes (d) inhibitors

Answer: (d) inhibitors — They prevent the active site from binding the substrate.

78. Competitive inhibitors have the same shape as the:

(a) substrate (b) enzyme
(c) product (d) co-enzyme

Answer: (a) substrate — They compete for the active site.

79. Competitive inhibition is:

(a) permanent (b) irreversible
(c) reversible (d) always fatal

Answer: (c) reversible — Stated in the book.

80. Malonate inhibits succinate dehydrogenase because it:

(a) is a protein (b) destroys the enzyme
(c) is a strong acid (d) resembles the substrate

Answer: (d) resembles the substrate — A competitive inhibitor.

81. Non-competitive inhibitors:

(a) change the shape of the active site
(b) have the same shape as the substrate
(c) increase enzyme activity
(d) act as substrates

Answer: (a) change the shape of the active site — The substrate can no longer bind.

82. Penicillin blocks the enzyme:

(a) carbonic anhydrase (b) trans-peptidase
(c) pepsin (d) succinate dehydrogenase

Answer: (b) trans-peptidase — Needed for the bacterial cell wall.

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83. Penicillin kills bacteria by disrupting the:

(a) ribosome (b) nucleus
(c) cell wall (d) flagellum

Answer: (c) cell wall — It is an irreversible inhibitor.

84. The enzyme in cobra venom is:

(a) PLA2 (b) pepsin
(c) amylase (d) catalase

Answer: (a) PLA2 — It disrupts red blood cells.

85. Destruction of red blood cells by cobra venom is called:

(a) hyperglycemia (b) hydrolysis
(c) hibernation (d) hemolysis

Answer: (d) hemolysis — It causes swelling and death.

27.4 LIPIDS

86. Lipids are:

(a) hydrophobic but soluble in organic solvents
(b) soluble in water
(c) polymers of glucose
(d) ionic salts

Answer: (a) hydrophobic but soluble in organic solvents — They are non-polar.

87. Unlike proteins and polysaccharides, lipids are not:

(a) sources of energy (b) organic
(c) found in cells (d) polymers

Answer: (d) polymers — They lack a repeating monomer unit.

88. Simple lipids are esters of fatty acids with:

(a) amines (b) alcohols
(c) phosphates only (d) sugars only

Answer: (b) alcohols — Fats, oils and waxes.

89. A triglyceride is formed from one glycerol and:

(a) one fatty acid (b) two fatty acids
(c) three fatty acids (d) three amino acids

Answer: (c) three fatty acids — Through three ester links.

90. Lipids which are solid at room temperature are called:

(a) liposomes (b) oils
(c) terpenes (d) fats

Answer: (d) fats — They contain saturated fatty acids.

91. Oils contain:

(a) unsaturated fatty acids (b) saturated fatty acids only
(c) no fatty acids (d) amino acids

Answer: (a) unsaturated fatty acids — With one or more double bonds.

92. Waxes are esters of long-chain fatty acids with:

(a) glycerol (b) long-chain alcohols
(c) ethanol (d) sterols

Answer: (b) long-chain alcohols — Both chains are non-polar.

93. Waxes form protective coatings because they are:

(a) ionic (b) hydrophilic
(c) hydrophobic (d) acidic

Answer: (c) hydrophobic — Water does not stick to the surface.

94. Which one is a natural wax?

(a) cholesterol (b) beeswax
(c) limonene (d) glycogen

Answer: (b) beeswax — Listed in the book.

95. Compound lipids contain an additional group such as:

(a) sulfonic acid (b) benzene
(c) chlorine (d) phosphate

Answer: (d) phosphate — Or a nitrogenous base, carbohydrate or protein.

96. A phospholipid has attached to glycerol:

(a) one fatty acid only
(b) three fatty acids
(c) two fatty acids and a phosphate group
(d) three phosphate groups

Answer: (c) two fatty acids and a phosphate group — The phosphate makes it polar.

97. The head of a phospholipid is:

(a) hydrophilic (b) hydrophobic
(c) non-polar (d) a hydrocarbon

Answer: (a) hydrophilic — It has an affinity for water.

98. In water, phospholipids form a:

(a) pleated sheet (b) helix
(c) bilayer (d) crystal lattice

Answer: (c) bilayer — A double-layered sheet.

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99. Lipoproteins transport in the bloodstream:

(a) amino acids (b) oxygen
(c) glucose (d) cholesterol and triglycerides

Answer: (d) cholesterol and triglycerides — Hydrophobic lipids.

100. Glycolipids are mostly found in:

(a) bone (b) nervous tissue
(c) hair (d) blood plasma

Answer: (b) nervous tissue — Lipids with sugar groups attached.

101. Liposomes are used for:

(a) hydrophilic drug delivery (b) making soap
(c) storing iron (d) building cell walls

Answer: (a) hydrophilic drug delivery — A lipid bilayer around a watery core.

102. Derived lipids are the:

(a) salts of minerals
(b) esters of glycerol
(c) polymers of fatty acids
(d) hydrolytic products of simple and compound lipids

Answer: (d) hydrolytic products of simple and compound lipids — E.g. fatty acids, glycerol, steroids.

103. The steroid nucleus consists of:

(a) two benzene rings
(b) four cyclopentane rings
(c) three cyclohexane rings and one cyclopentane ring
(d) one cyclohexane ring only

Answer: (c) three cyclohexane rings and one cyclopentane ring — Rings A, B, C and D.

104. Steroids with an –OH group are called:

(a) terpenes (b) sterols
(c) waxes (d) glycolipids

Answer: (b) sterols — They belong to the alcohol class.

105. Which one is a steroid?

(a) cholesterol (b) limonene
(c) beeswax (d) glycerol

Answer: (a) cholesterol — Also bile salts, vitamin D and steroid hormones.

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106. Terpenes are built from units of:

(a) amino acids (b) glucose
(c) glycerol (d) isoprene

Answer: (d) isoprene — Hydrocarbons of plants.

107. Terpenes containing 40 carbon atoms are called:

(a) carotenoids (b) sterols
(c) waxes (d) phospholipids

Answer: (a) carotenoids — Stated in the book.

108. Limonene is a terpene found in:

(a) milk (b) lemon
(c) liver (d) honey

Answer: (b) lemon — Stated in the book.

109. The fat-soluble vitamins are:

(a) A and C only (b) B and C
(c) A, D, E and K (d) B, C and K

Answer: (c) A, D, E and K — Lipids help in their absorption.

110. Compared with carbohydrates, lipids provide per gram:

(a) no energy (b) less energy
(c) equal energy (d) more energy

Answer: (d) more energy — They have a higher caloric value.

111. Fats insulate the body due to their:

(a) ionic nature (b) non-polar nature
(c) acidity (d) high solubility in water

Answer: (b) non-polar nature — From water, heat and electricity.

112. Hibernating animals store extra:

(a) protein (b) cellulose
(c) fat (d) minerals

Answer: (c) fat — A more concentrated source of energy than glycogen.

27.5 NUCLEIC ACIDS

113. Nucleic acids are polymers of:

(a) nucleotides (b) amino acids
(c) monosaccharides (d) fatty acids

Answer: (a) nucleotides — They store and transmit genetic information.

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114. The sugar in RNA is:

(a) glucose (b) deoxyribose
(c) ribose (d) fructose

Answer: (c) ribose — A five-membered sugar.

115. The sugar in DNA is:

(a) ribose (b) deoxyribose
(c) sucrose (d) maltose

Answer: (b) deoxyribose — A five-membered sugar.

116. Nucleotides are joined by:

(a) hydrogen bonds (b) peptide linkages
(c) glycosidic linkages (d) phosphodiester linkages

Answer: (d) phosphodiester linkages — Between the 5′ and 3′ carbons of the sugars.

117. Phosphodiester linkages are:

(a) ionic bonds (b) weak hydrogen bonds
(c) strong covalent bonds (d) van der Waals forces

Answer: (c) strong covalent bonds — They give stability to the chain.

118. A section of DNA which makes a specific protein is a:

(a) codon (b) gene
(c) ribosome (d) nucleotide

Answer: (b) gene — Each protein is made by a specific gene.

119. Humans have chromosomes numbering:

(a) 23 pairs (b) 23
(c) 46 pairs (d) 12 pairs

Answer: (a) 23 pairs — Received from the parents.

120. A group of three bases coding for a specific amino acid is a:

(a) genome (b) gene
(c) chromosome (d) codon

Answer: (d) codon — Shown in the book's figure.

121. The double helix structure of DNA was shown by:

(a) Watson and Crick (b) Emil Fischer
(c) Ziegler and Natta (d) Robinson

Answer: (a) Watson and Crick — In 1953.

122. The double helix of DNA was proposed in:

(a) 1903 (b) 1894
(c) 1972 (d) 1953

Answer: (d) 1953 — By James Watson and Francis Crick.

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123. In DNA, adenine always pairs with:

(a) cytosine (b) guanine
(c) thymine (d) uracil

Answer: (c) thymine — By two hydrogen bonds.

124. In DNA, guanine always pairs with:

(a) adenine (b) cytosine
(c) thymine (d) uracil

Answer: (b) cytosine — By three hydrogen bonds.

125. The number of hydrogen bonds between A and T is:

(a) three (b) one
(c) two (d) four

Answer: (c) two — G–C has three.

126. The number of hydrogen bonds between G and C is:

(a) three (b) one
(c) two (d) four

Answer: (a) three — A–T has two.

127. The two chains of DNA run:

(a) in a circle (b) in the same direction
(c) at right angles (d) opposite to each other

Answer: (d) opposite to each other — The 3′ end of one faces the 5′ end of the other.

128. The copying of DNA during the cell cycle is called:

(a) translation (b) replication
(c) transcription (d) denaturation

Answer: (b) replication — Each new cell gets a complete set.

129. RNA is usually:

(a) triple stranded (b) double stranded
(c) single stranded (d) a double helix

Answer: (c) single stranded — It can fold back into a helical loop.

130. The base found in RNA in place of thymine is:

(a) uracil (b) adenine
(c) guanine (d) cytosine

Answer: (a) uracil — RNA bases: A, U, G, C.

131. The RNA which carries the genetic blueprint from DNA is:

(a) none of them (b) tRNA
(c) rRNA (d) mRNA

Answer: (d) mRNA — Messenger RNA.

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132. The RNA which helps to make up the ribosome is:

(a) mRNA (b) rRNA
(c) tRNA (d) DNA

Answer: (b) rRNA — Ribosomal RNA.

133. The RNA which translates the base sequence into amino acids is:

(a) rRNA (b) mRNA
(c) tRNA (d) DNA

Answer: (c) tRNA — Transfer RNA.

134. Copying a DNA gene into mRNA is called:

(a) translation (b) transcription
(c) replication (d) hydrolysis

Answer: (b) transcription — It takes place in the nucleus.

135. Conversion of the mRNA code into a sequence of amino acids is:

(a) translation (b) transcription
(c) replication (d) inhibition

Answer: (a) translation — It takes place at the ribosomes.

136. Translation takes place at the:

(a) chromosomes (b) nucleus
(c) cell wall (d) ribosomes

Answer: (d) ribosomes — In the cytoplasm.

137. The DNA strand opposite to A–T–G–C is:

(a) U–A–C–G (b) A–T–G–C
(c) T–A–C–G (d) G–C–A–T

Answer: (c) T–A–C–G — A pairs with T and G with C.

27.5 MINERALS

138. Minerals are:

(a) carbohydrates (b) proteins
(c) lipids (d) inorganic substances

Answer: (d) inorganic substances — Needed for life processes.

139. Minerals:

(a) produce no energy (b) are the main fuel
(c) are polymers (d) are hormones

Answer: (a) produce no energy — But they have important functions.

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140. Iron is needed for the synthesis of:

(a) insulin (b) haemoglobin
(c) keratin (d) cellulose

Answer: (b) haemoglobin — And myoglobin in muscle cells.

141. Myoglobin is found in:

(a) bone (b) red blood cells
(c) muscle cells (d) hair

Answer: (c) muscle cells — It contains iron.

142. A good source of iron is:

(a) liver (b) salt
(c) milk (d) fruit juice

Answer: (a) liver — Also meat, egg yolks and lentils.

143. The mineral needed for normal blood clotting is:

(a) zinc (b) iron
(c) calcium (d) sodium

Answer: (c) calcium — Also for bones and teeth.

144. A good source of calcium is:

(a) fruit juices (b) salt
(c) whole grains only (d) milk and dairy products

Answer: (d) milk and dairy products — Also greens and sardines.

145. The mineral which regulates the release and use of body energy (ATP) is:

(a) iron (b) phosphorus
(c) zinc (d) calcium

Answer: (b) phosphorus — Stated in Table 27.1.

146. Phosphorus helps to carry fat in the body as part of:

(a) keratin (b) glycogen
(c) phospholipids (d) haemoglobin

Answer: (c) phospholipids — Stated in Table 27.1.

147. The mineral found in the hormone insulin is:

(a) sodium (b) iron
(c) calcium (d) zinc

Answer: (d) zinc — It is also part of many enzyme systems.

148. Sodium, potassium and chloride maintain:

(a) fluid balance (b) blood clotting
(c) oxygen transport (d) bone formation

Answer: (a) fluid balance — Also nerve transmission and muscle function.

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149. Macro-minerals are required in amounts:

(a) less than 100 mg/dL (b) greater than 100 mg/dL
(c) less than 1 mg/dL (d) exactly 10 mg/dL

Answer: (b) greater than 100 mg/dL — Stated in the book.

150. Which one is a trace (micro) element?

(a) sodium (b) calcium
(c) phosphorus (d) iodine

Answer: (d) iodine — According to the book's list.

151. Phosphorus and calcium both aid in the formation of:

(a) haemoglobin (b) bones and teeth
(c) insulin (d) glycogen

Answer: (b) bones and teeth — Stated in Table 27.1.

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